

Introduction to Neural Networks

Transposed convolution

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April 13, 2026©



Motivation

Reduction of the feature map

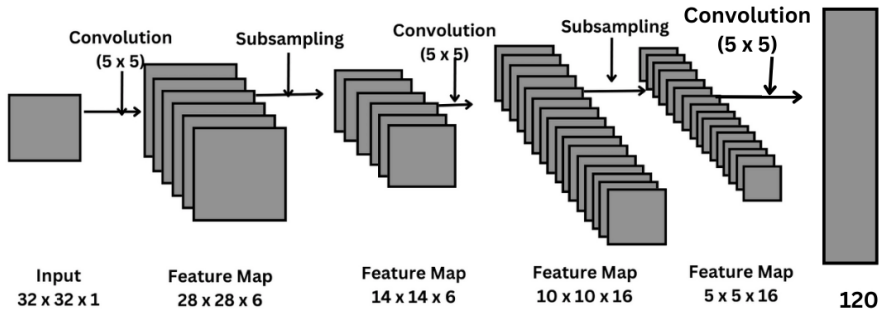


Figure: Convolutional part of LeNet5

- Given that

$$G = I \star K$$

- Is this possible?

$$I = G \star K^{-1}$$

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- Is this possible?

$$I = G \star K^{-1}$$

- In theory, but it is not practical, it involves the computation of a inverse matrix.
- Therefore we will apply a transposed convolution instead.

BENJAMIN BUTTON'S 10 YEAR CHALLENGE



2009

2019



Transposed convolution

- Reverse the dimension of the convolution operation, upsampling

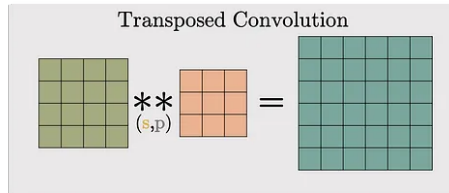
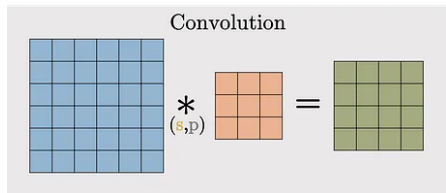
$$G = I \star K$$

$$T = G \star \star W$$

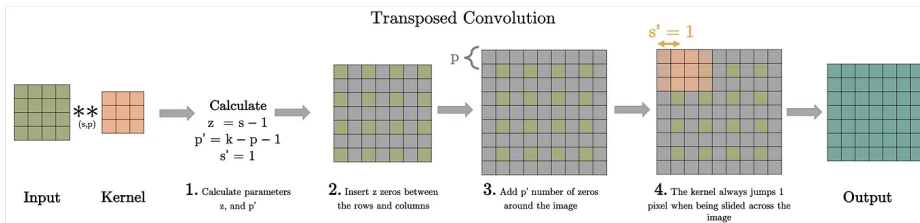
$$|T| = |I|$$

$$T \approx I$$

- Transposed convolution doesn't reverse the standard convolution by values



Transposed convolution



- ① Calculate parameters z and p' :

$$z = s - 1 \quad (1)$$

$$p' = k - p - 1 \quad (2)$$

- ② Between each row and columns of the input, G , insert z number of zeros.
- ③ Pad the modified input image with p' number of zeros
- ④ Perform a convolution, $G \star W$, with stride $s' = 1$

- Dumoulin, Vincent, and Francesco Visin. "A guide to convolution arithmetic for deep learning." arXiv preprint arXiv:1603.07285 (2016).
- Aqeel Anwar, What is Transposed Convolutional Layer?, <https://towardsdatascience.com/what-is-transposed-convolutional-layer>, Consultado: Abril 2023.